TrES-5b

R-0752

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_241003` · created 2026-07-28T03:01:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460586.6547 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.028
Transit depth (Rp/Rs)^2	2.29 +/- 0.84 [%]
Orbital Inclination (inc)	88.98 +/- 1.87
Airmass coefficient 1 (a1)	1.031 +/- 0.027
Airmass coefficient 2 (a2)	-0.02 +/- 0.1
Scatter in the residuals of the lightcurve fit is	2.62 %
Best Comparison Star	#1 - [84, 96]
Optimal Aperture	3.12
Optimal Annulus	9.55
Transit Duration (day)	0.0865 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.028 vs 0.142 (deviation 0.26σ)
Duration fit vs published	0.0865 d vs 0.0758 d (deviation 2.6σ)
Mid-time O–C	8.3 min
Depth SNR	4.3
Residual scatter	2.62 %

Light curve

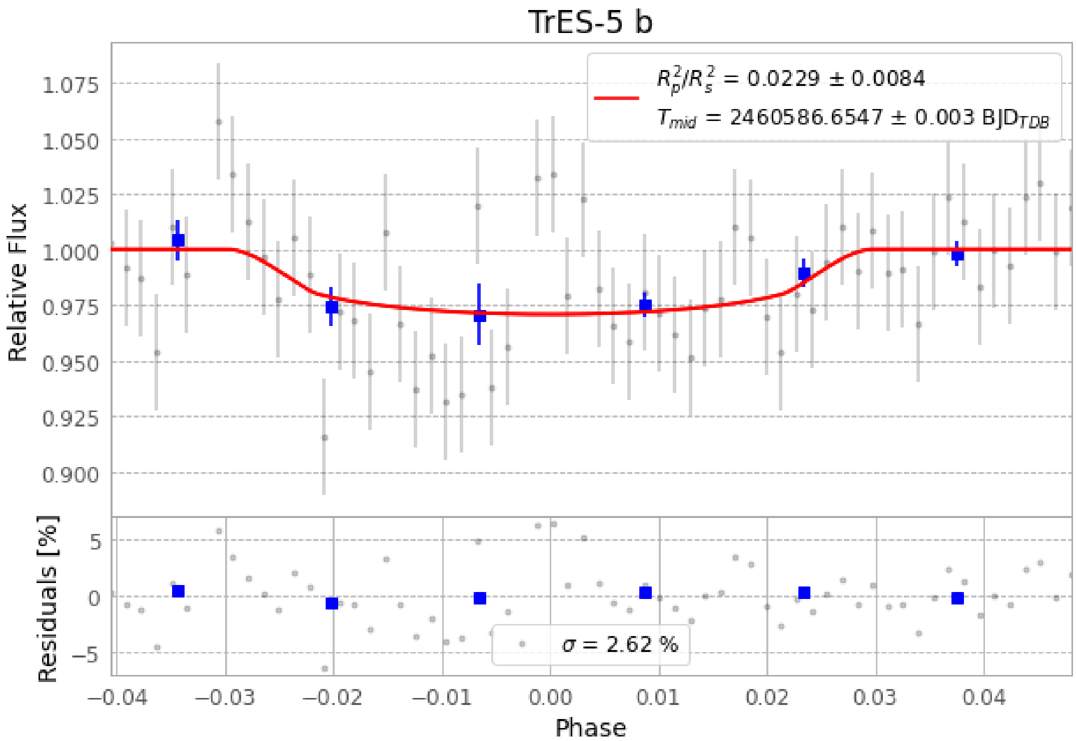

Figure 1 — FinalLightCurve_TrES-5 b_2024-10-02.png (SHA-256 defe4f5929bd3dfa…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 244], 1 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a291ead952e1b184…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

64 FITS frames (42 MB), `TRES-5241003021215.FITS` → `TRES-5241003052115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-241003.log (7e3159fe7400…), FinalLightCurve_TrES-5 b_2024-10-02.png (defe4f5929bd…), FinalLightCurve_TrES-5 b_2024-10-02.csv (2f45fee84abc…)

Compute resources

Wall time 75.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)